

Introduction

Bartlett's test evaluates homogeneity of variance across k groups. It is the likelihood-ratio test under normality, and it is the most powerful variance-homogeneity test when normality holds. Its major drawback: it is extremely sensitive to non-normality, losing validity as soon as the normality assumption is relaxed.

Prerequisites

Variance, ANOVA assumptions.

Theory

For k groups with sample sizes n_i and sample variances s_i^2 , let $N = \sum n_i$ and pooled variance $s_p^2 = \frac{\sum (n_i - 1)s_i^2}{N - k}$. The test statistic is

$$K = \frac{(N - k) \ln s_p^2 - \sum (n_i - 1) \ln s_i^2}{1 + \frac{1}{3(k-1)} \left(\sum \frac{1}{n_i - 1} - \frac{1}{N - k} \right)} \sim \chi_{k-1}^2 \text{ under } H_0.$$

Rejecting H_0 indicates variances differ; failing to reject suggests equality.

Assumptions

- Independent observations within and across groups.
- **Normality of within-group distributions** – strongly required.
- Moderate sample sizes (≥ 5 per group).

R Implementation

```
set.seed(2026)

df <- data.frame(
  group = factor(rep(c("A", "B", "C"), each = 40)),
  y      = c(rnorm(40, 50, 5),
             rnorm(40, 55, 5),
             rnorm(40, 52, 5))
)

bartlett.test(y ~ group, data = df)

# Under normality, Bartlett and Levene should agree
library(car)
leveneTest(y ~ group, data = df)

# Under non-normality: Bartlett over-rejects
df2 <- df
df2$y <- c(rexp(40, rate = 1/50),
          rexp(40, rate = 1/55),
          rexp(40, rate = 1/52))
bartlett.test(y ~ group, data = df2)
leveneTest(y ~ group, data = df2)
```

Output & Results

Under normality:

```
Bartlett's K-squared = 0.08, df = 2, p-value = 0.96
Levene's F = 0.10, p-value = 0.91
```

Under exponential:

```
Bartlett's K-squared = 8.7, df = 2, p-value = 0.013    # false positive
Levene's F = 1.5, p-value = 0.24                      # correctly holds
```

Bartlett falsely rejects equal variances under non-normal data. Levene correctly maintains its Type I rate.

Interpretation

Almost never use Bartlett in practice; prefer Levene (median-based) for robustness. Report Bartlett only when normality is firmly established (e.g., by design or by a prior transformation).

Practical Tips

- Bartlett is theoretically optimal under normality but rarely appropriate for real data.
- Avoid the Bartlett-then-ANOVA pipeline; use Welch's ANOVA directly.
- For skewed or heavy-tailed data, Levene or the Fligner-Killeen test are safer.
- The null hypothesis “all variances are equal” is rarely the clinical question; it is a diagnostic step for another test.
- Many software packages report Bartlett as default; override to Levene for applied work.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests